Compatibility policy

This policy defines how long BackupSage's public contracts remain compatible and how a future breaking change is announced. It applies to released v1.x versions; unreleased main-branch behavior is not a compatibility promise.

Window and change rules

Within v1.x, every surface listed below is stable. Additions are permitted where stated; removals, renames, semantic changes, and incompatible format changes are not. A v1.x release must continue to read the on-disk data formats created by earlier v1.x releases.

A breaking change requires a new major version. Before that major release, BackupSage will document the affected surface, migration or re-indexing steps, and the first release containing the change in both this policy and its release notes. A deprecation notice may be published during v1.x, but it does not make the v1.x contract breakable.

Public surfaces

Surface	Current form	v1.x promise
JSON reports	`dedup --json` report `"version": 1`; `search --json`, `search --all --json`, `master list --json`, and `master verify --json`	Stable and additive-only. Existing keys retain their names, types, and meanings; `dedup` remains report version 1. v1.0.2 added the keeper-star fields (#9): `actionable`, `review_only_bytes`, `transitive_only_files`, `actionable_rule` — additive. In the same release `reclaimable_bytes`/`duplicate_files` stopped counting transitive-only members (a correctness fix to match their documented meaning: only members measured safe against the keeper are reclaimable). v1.0.2 also added `search --json`'s top-level `mode` (#70) — additive; `matches` is `null` only on indexes built with the new `--mode search-only`, and output for `--mode full` (the default, and every index built by an earlier v1.x) is unchanged.
Exit codes	`0` clean, `1` error, `2` completed with skips; clap usage errors currently also use `2`	Stable meanings, including the documented usage-error caveat.
Per-source index	SQLite schema version `3`	Stable within v1.x. v1.x continues to read schema-v3 indexes and the supported schema-v2 indexes, which remain `v2-limited` where metadata required by a command is absent.
Master catalog	SQLite master catalog containing registered index metadata and replicas	Stable within v1.x. A catalog created by an earlier v1.x release remains readable by later v1.x releases; registration and query behavior preserve existing entries.
Perceptual hash	`sage-dct-v1` 64-bit DCT pHash	Frozen. The algorithm identifier and its hash semantics do not change; any future algorithm receives a new identifier, and mixed algorithms are not compared.
CLI	Documented commands, subcommands, flags, defaults, and accepted values	Stable within v1.x. New commands or flags may be added, but existing invocation syntax and behavior are not removed or repurposed. `dedup --threshold` remains capped at `0..3`, because its four 16-bit bands guarantee recall only through Hamming distance 3.

`docs/CONTRACT.md` records the exact current JSON and exit-code shapes and the golden tests that enforce them. This policy adds the version window and deprecation procedure; it does not widen a surface that is documented there as human-oriented and unstable.

On-disk data

BackupSage never rewrites an existing index or master catalog merely to read it. A later v1.x release may create a newer index only through the normal explicit indexing flow. If a future major version cannot read an old format, it will provide the migration or re-indexing instructions required by the deprecation procedure above before that breaking release ships.